TAI BUI

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PROFESSIONAL SUMMARY

Master of Science in Computer Science candidate (December 2026) with production experience building applied large language model (LLM) tooling — Retrieval-Augmented Generation (RAG) pipelines, agent orchestration, and Model Context Protocol (MCP) server integrations that cut website build time 65% across 170+ sites. Deep learning background in PyTorch, with an emphasis on rigorous model evaluation. Seeking Machine Learning / AI Engineer roles beginning January 2027.

SKILLS

Languages: Python, JavaScript, C++, SQL, PHP, HTML/CSS

Machine Learning & AI: PyTorch, TensorFlow, Keras, Scikit-learn, Retrieval-Augmented Generation (RAG), Model Context Protocol (MCP), LlamaIndex, FAISS, LLaMA 3.1, transformers, variational autoencoders (VAE), agent orchestration, model evaluation

Tools & Platforms: Git, GitHub Actions, CI/CD, Amazon Web Services (AWS), Linux, React, WordPress; working exposure to Docker, TypeScript, and REST APIs

WORK EXPERIENCE

Web Developer — EOS Healthcare Marketing
March 2026 – Present

- Build applied LLM tooling adopted as the team standard — Claude skills backed by Model Context Protocol (MCP) servers, plus baseline theme automation — reducing average website build time from 90 hours to under 30, a 65% reduction.
- Maintain a portfolio of 170+ production healthcare websites as one of two developers, delivering new builds and owning ongoing debugging, maintenance, and security updates.
- Automate local-to-staging deployments and uptime monitoring in GitHub Actions — the team's first CI/CD pipeline — with alerting that automatically files an issue when a site goes down.

Teaching Assistant, Algorithms — University of Colorado Denver
January 2024 – May 2024

- Taught sorting, graph algorithms, and complexity analysis to a class of 30 students through weekly office hours and one-on-one tutoring.
- Graded assignments and delivered written feedback; students attending regular tutoring showed measurable exam improvement.

PROJECTS

Multimodal Disease Subtyping (TCGA-BRCA) — Team of 4
January 2026 – May 2026

Python, PyTorch, variational autoencoders, Scikit-learn

- Owned the RNA sequencing variational autoencoder (VAE), learning a latent representation of gene expression profiles for breast cancer subtype classification and reaching 0.63 accuracy against a 0.58 baseline.
- Benchmarked 4 fusion strategies (early concatenation, contrastive latent alignment, joint multimodal VAE, raw early fusion) against unimodal baselines; established that none surpassed the clinical-only VAE (0.68 accuracy, 0.59 weighted F1), tracing the cause to weak cross-modal alignment introducing noise rather than signal.
- Evaluated all 8 models on balanced accuracy alongside raw accuracy, exposing majority-class collapse in the RNA-only models (0.25 balanced accuracy) that headline accuracy had masked, and redirecting the team's approach.

Health Insurance Policy Assistant
January 2026 – May 2026

Python, LLaMA 3.1, LlamaIndex, FAISS

- Built a Retrieval-Augmented Generation (RAG) assistant that orchestrates subagents to retrieve, summarize, and compare plan coverage details from a structured policy dataset, grounding responses in source records to prevent hallucinated coverage claims.
- Implemented the vector search and retrieval layer in FAISS and LlamaIndex, iterating on chunking strategy and prompt design to improve groundedness.

CERTIFICATIONS

AWS Certified Cloud Practitioner | NVIDIA: Building Transformer-Based Natural Language Processing (NLP) Applications | Software Engineering Certificate, University of Colorado Denver

EDUCATION

Master of Science, Computer Science — University of Colorado Denver
Expected December 2026

Coursework: Machine Learning, Deep Learning, Generative AI, Big Data Science, Advanced Computer Architecture

Bachelor of Arts, Computer Science (Biology Minor) — University of Colorado Denver May 2024
GPA: 3.87 / 4.00